Bitcoin Energy Consumption and Price Clustering Analysis with Causality Testing

Abstract

This paper analyzes the relationship between Bitcoin energy consumption, ESG index, carbon emissions, and Bitcoin price. We apply K-Means clustering to identify potential market regimes and conduct Granger causality tests to examine whether energy consumption predicts Bitcoin price fluctuations.

1 Introduction

The increasing energy consumption of Bitcoin mining has raised concerns regarding its environmental impact. This study aims to analyze Bitcoin's energy consumption, ESG index, carbon emissions, and price fluctuations using machine learning and econometric techniques.

Several previous studies have investigated Bitcoin price fluctuations and energy consumption. For instance, Stoll et al. (2019) analyzed the carbon footprint of Bitcoin mining, while Bouri et al. (2019) examined its volatility characteristics. Additionally, Diebold Yilmaz (2020) provided insights into financial risk related to Bitcoin markets.

2 Data Processing

2.1 Data Sources

The datasets used include:

- Bitcoin and Ethereum energy consumption
- Bitcoin price (USD)
- ESG stock index
- Carbon emissions ($MtCO_2e$)

2.2 Preprocessing Steps

1. Standardized date formats and set as the index.
2. Adjusted column names for clarity.
3. Selected relevant features such as energy consumption, Bitcoin price, and ESG index.
4. Converted price data to numeric format and removed missing values.
5. Merged all datasets using the date index.

3 Clustering Analysis

3.1 Feature Selection

The selected features for clustering analysis include:

- Estimated Bitcoin energy consumption (TWh/year)
- Minimum Bitcoin energy consumption (TWh/year)
- Annual global energy consumption (TWh)
- S&P World ESG Index (USD)
- Bitcoin carbon emissions ($MtCO_2e$)
- Bitcoin price (USD)

3.2 Standardization

Features are standardized using z-score normalization:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma} \tag{1}$$

where μ is the mean and σ is the standard deviation.

3.3 Elbow Method for Optimal K

The sum of squared errors (SSE) is computed for cluster values ranging from $K = 2$ to $K = 10$, and the optimal K is selected at the point where the SSE curve flattens.

$$\text{SSE} = \sum_{i=1}^{K} \sum_{j \in C_i} ||X_j - \mu_i||^2 \tag{2}$$

where μ_i is the centroid of cluster C_i.

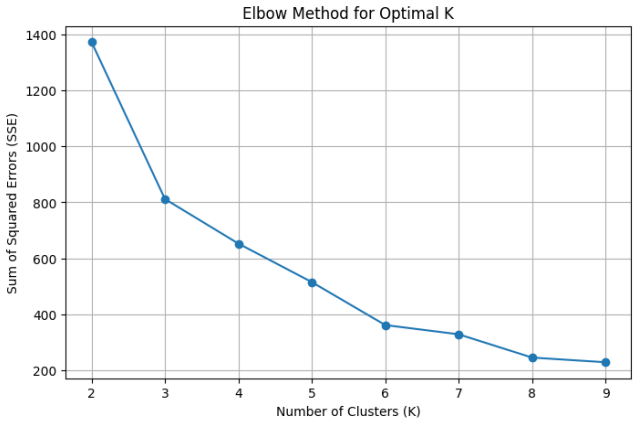


Figure 1: Elbow Method for Optimal K

3.4 K-Means Clustering

K-Means clustering is applied, and data points are assigned to clusters:

$$\mu_i = \frac{1}{|C_i|} \sum_{j \in C_i} X_j \tag{3}$$

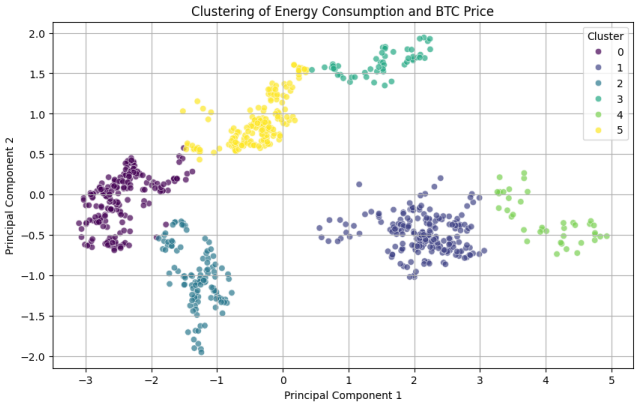


Figure 2: Cluster Visualization using PCA

3.5 Silhouette Score

The clustering performance is evaluated using the silhouette score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \tag{4}$$

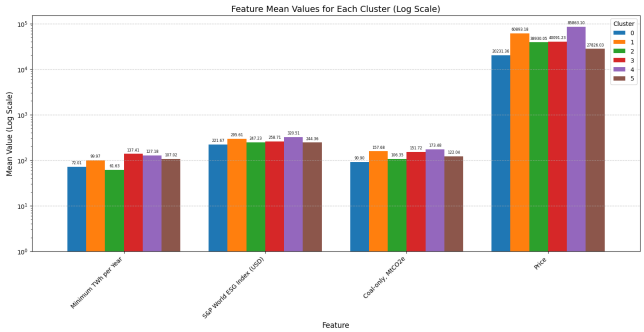


Figure 3: Cluster Analysis

where $a(i)$ is the average intra-cluster distance and $b(i)$ is the average inter-cluster distance.

Silhouette Score: **0.5249**

4 Granger Causality Test

The test evaluates whether Bitcoin energy consumption Granger-causes Bitcoin price. The null hypothesis is:

$$H_0 : \text{Energy consumption does not Granger-cause Bitcoin price} \tag{5}$$

Table 1: Granger Causality Test Results

Lags	F-Statistic	p-value
1	11.29	0.0008
2	4.40	0.0125
3	2.87	0.0354
4	2.14	0.0731
5	1.71	0.1279

5 Conclusion

This study provides insights into the relationship between Bitcoin energy consumption and price fluctuations. The clustering analysis identifies distinct market regimes, while the Granger causality test suggests that Bitcoin energy consumption has short-term predictive power over its price.

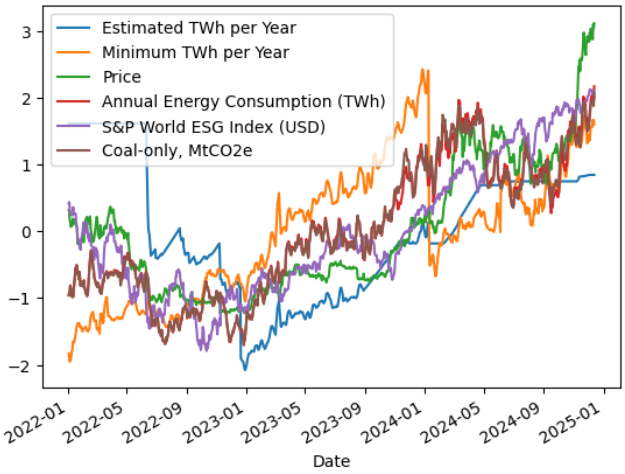


Figure 4: Bitcoin Price and Energy Consumption Trends

6 References

References

Stoll, C., Klaassen, L., Gallersdorfer, U., Neumann, D. (2019). The Carbon Footprint of Bitcoin. Joule, 3(7), 1647-1661.

Bouri, E., Jain, A., Roubaud, D., Kristoufek, L. (2019). Modelling the volatility of Bitcoin: An empirical study. Finance Research Letters, 29, 282-287.

Diebold, F. X., Yilmaz, K. (2020). Measuring Financial Asset Return and Volatility Spillovers, with Application to Global Equity Markets. The Economic Journal, 120(544), 1583-1611.